

Macro Nexus Forecast(rewrite)

June 8, 2026

1 Forecasting Macroeconomic Trends: The Nexus of Exchange Rates, Energy, and CP

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRFRegressor
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.statespace.sarimax import SARIMAX
import yfinance as yf
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import adfuller
from pmdarima.arima.utils import nsdiffs
from pmdarima import auto_arima
from sklearn.metrics import mean_squared_error, mean_absolute_error, \
    root_mean_squared_error
import warnings
warnings.filterwarnings('ignore', category=FutureWarning)
from xgboost import XGBRegressor
```

1.1 1. Data Collection

1.1.1 1.1 Inflation

```
[2]: # Inflation is pulled from bsp site.
# link: https://www.bsp.gov.ph/Statistics/Prices/tab34_inf_2018.aspx

inflation_df = pd.read_excel('Inflation_rates_dataset_2021-2026.xlsx')
inflation_df['month'] = pd.to_datetime(inflation_df['month']).dt.to_period('M').\
    dt.to_timestamp()
inflation_df = inflation_df.set_index('month')
```

1.1.2 1.2 Sentiment Analysis

```
[3]: # Sentiment Analysis was scraped, cleaned, process in different .ipynb as it
      ↳ takes time to scrape headers
      # Newslime headers source link: www.bworldonline.com/banking-finance/

      sentiment_df = pd.read_csv('finBERT Sentiment-Inflation.csv')
      sentiment_df['month'] = pd.to_datetime(sentiment_df['date']).dt.to_period('M').
      ↳ dt.to_timestamp()
      sentiment_df = sentiment_df.drop(columns = ['date'], errors = 'ignore')
```

1.1.3 1.3 Oil Brent

```
[4]: oil_df = yf.download(tickers='BZ=F', start='2021-01-01', end = '2026-06-01',
      ↳ interval='1d')
      oil_df = oil_df['Close'].resample('M').mean()
      oil_df.columns = ['oil_brent']
      oil_df = oil_df.reset_index()
      oil_df['month'] = oil_df['Date']
      oil_df = oil_df.drop(columns = 'Date', errors = 'ignore')
      oil_df['month'] = pd.to_datetime(oil_df['month']).dt.to_period('M').dt.
      ↳ to_timestamp()
```

[*****100%*****] 1 of 1 completed

1.1.4 1.4 Merging dataframes

```
[5]: df = pd.merge(sentiment_df, oil_df, on='month', how='inner').drop(columns =
      ↳ 'inflation', errors='ignore').set_index('month')
      df = df.join(inflation_df, how = 'left')
```

```
[6]: print(f'DataFrame Information:')
      print(f'Length: {len(df)}')
      print(f'NAs: {df.isna().sum()}')
      print('Head:')
      print(df.head())
      print('Tail:')
      print(df.tail())
```

```
DataFrame Information:
Length: 65
NAs: sentiment      0
      oil_brent      0
      inflation      2
dtype: int64
Head:
      sentiment  oil_brent  inflation
month
```

2021-01-01	-0.096034	55.351579	3.7
2021-02-01	0.019575	62.227894	4.2
2021-03-01	-0.041243	65.702174	4.1
2021-04-01	0.023667	65.328572	4.1
2021-05-01	-0.017046	68.258500	4.1

Tail:

	sentiment	oil_brent	inflation
month			
2026-01-01	-0.028967	64.765499	2.0
2026-02-01	-0.000079	69.406843	2.4
2026-03-01	0.023276	99.599546	4.1
2026-04-01	0.013479	102.464286	NaN
2026-05-01	0.078773	104.092000	NaN

1.2 2 Exploratory Data Analysis

1.2.1 2.1 Correlation Analysis and Stationarity Check

```
[7]: print('Energy vs Inflation correlation through lags:')
print('')
for lag in [0, 1, 2, 3, 4]:
    lagged_corr = df['inflation'].corr(df['oil_brent'].rolling(window = 3).
    ↪mean().shift(lag))
    print(f"Oil Brent (Lag {lag}) vs Inflation Corr: {lagged_corr:.4f}")

print('')
print('Sentiment Analysis vs Inflation Correlation through lags:')
print('')
for lag in [0, 1, 2, 3, 4]:
    lagged_corr = df['inflation'].dropna().corr(df['sentiment'].rolling(window=
    ↪3).mean().shift(lag))
    print(f"Sentiment (Lag {lag}) vs Inflation Corr: {lagged_corr:.4f}")
print('')

print('ADF Test for stationarity checking')

def adf_test(series):
    result = adfuller(series)[1]

    if result > 0.05:
        print(f'ADF: {result}, differencing is needed')
    else:
        print(f'ADF: {result}, no differencing needed. Data is stationary')

def ndiffs(series):
    result = nsdiffs(series, m=12, test='ocsb')
    print(f'Suggested Seasonal Differencing: {result}')
adf_test(df['inflation'].dropna())
```

```
ndiffs(df['inflation'].dropna())
```

Energy vs Inflation correlation through lags:

```
Oil Brent (Lag 0) vs Inflation Corr: 0.6226
Oil Brent (Lag 1) vs Inflation Corr: 0.6493
Oil Brent (Lag 2) vs Inflation Corr: 0.6735
Oil Brent (Lag 3) vs Inflation Corr: 0.7006
Oil Brent (Lag 4) vs Inflation Corr: 0.7229
```

Sentiment Analysis vs Inflation Correlation through lags:

```
Sentiment (Lag 0) vs Inflation Corr: 0.4130
Sentiment (Lag 1) vs Inflation Corr: 0.4294
Sentiment (Lag 2) vs Inflation Corr: 0.4417
Sentiment (Lag 3) vs Inflation Corr: 0.4162
Sentiment (Lag 4) vs Inflation Corr: 0.3793
```

ADF Test for stationarity checking

ADF: 0.13346084422113258, differencing is needed

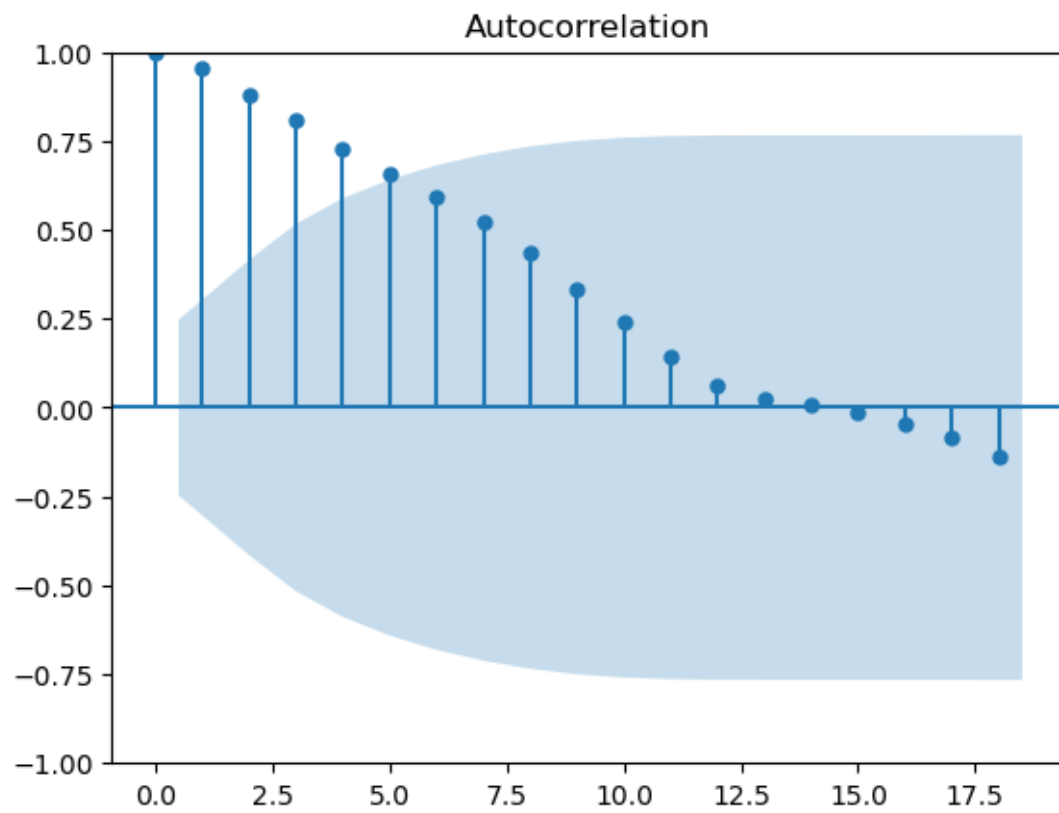
Suggested Seasonal Differencing: 0

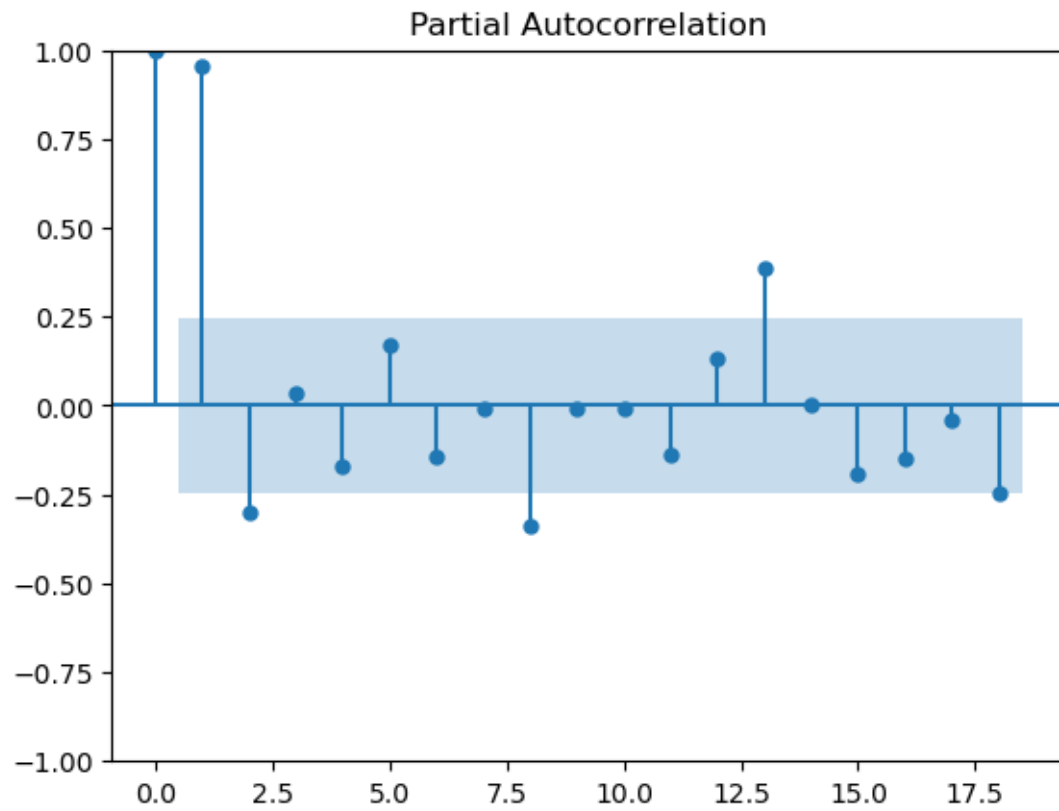
1.2.2 2.2 ACF/PACF Plots

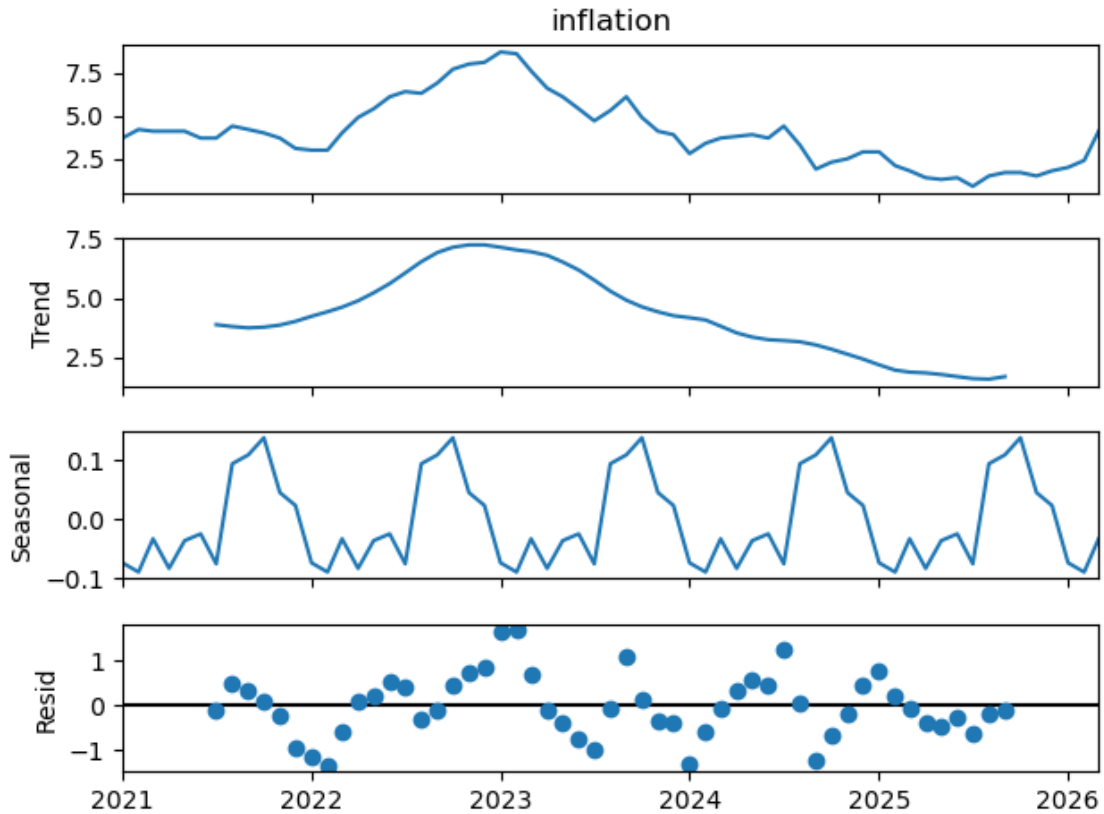
```
[8]: plot_acf(df['inflation'].dropna())
plot_pacf(df['inflation'].dropna())

decomp = seasonal_decompose(df['inflation'].dropna(), model = 'additive',
                             period = 12)
decomp.plot()
plt.tight_layout()
plt.show()

# This shows that nsdiffs prove that no Seasonal Differencing needed
# Other tests like ACF and PACF and Seasonal Decomposition shows that there is
↪ a weak seasonality.
# Auto ARIMA will be used and make sure that Seasonal Differencing is not
↪ included
```







1.3 3. Feature Engineering

1.3.1 3.1 Feature Engineering

```
[9]: df['oil_brent'] = df['oil_brent'].rolling(window=3).mean().shift(2)
df['sentiment'] = df['sentiment'].rolling(window=3).mean().shift(2)
df = df[5:]
X_future = df[-2:].drop(columns = 'inflation', errors='ignore')
```

1.3.2 3.2 Train/Test Split

```
[35]: split = int(len(df)*0.8)
train = df.iloc[:split]
test = df.iloc[split:]

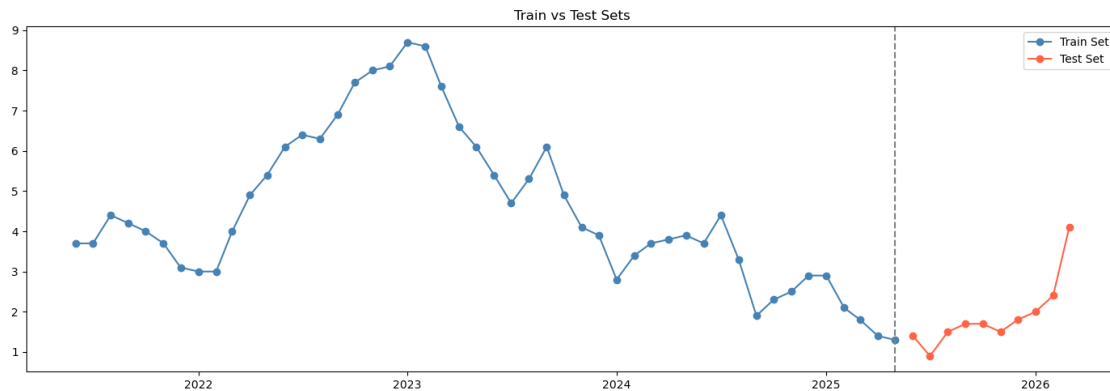
fig, ax = plt.subplots(figsize=(14,5))

ax.plot(train['inflation'], color = 'steelblue', label='Train Set', marker='o')
ax.plot(test['inflation'], color = 'tomato', label='Test Set',marker='o')

ax.axvline(train.index.max(), color='grey', linestyle='--')
```

```
ax.set_title('Train vs Test Sets')

plt.legend()
plt.tight_layout()
plt.show()
```



1.4 4. Modeling and Initial Evaluation

4.1 SARIMA Modeling

```
[11]: arima_model = auto_arima(train['inflation'].dropna(),
                                seasonal = True,      # Seasonality is weak but there's
                                ↪signal so we want to capture it
                                m=12,                  # For monthly data
                                d=1,                    # ADF test proved that
                                ↪differencing is needed
                                D=0,                    # nsdiff proved that seasonal
                                ↪differencing is not needed
                                max_p=3,
                                max_q=3,
                                max_P=2,
                                max_Q=2,
                                stepwise=False,        # For exhaustive search instead of
                                ↪stepwise
                                information_criterion='aic')
```

4.2 Model Summary, Forecast, Evaluation and Plot

```
[34]: # SARIMA model summary
print(arima_model.summary())

# Forecasting
forecast, conf_int = arima_model.predict(n_periods=len(test.dropna()),
    ↪return_conf_int=True)
```

```

# SARIMA Forecast Evaluation
def err_metrics(y_true, y_pred):
    # 1. Mean Squared Error (MSE)
    mse = mean_squared_error(y_true, y_pred)

    # 2. Root Mean Squared Error (RMSE)
    # In newer versions of sklearn, you can use squared=False
    rmse = root_mean_squared_error(y_true, y_pred)

    # 3. Mean Absolute Error (MAE)
    mae = mean_absolute_error(y_true, y_pred)
    return{
        print(f"MSE: {mse}"),
        print(f"RMSE: {rmse}"),
        print(f"MAE: {mae}")}

err_metrics(test['inflation'].dropna(), forecast.values)

# Forecasting Plot
forecast_df = pd.DataFrame({
    'forecast' : forecast.values,
    'lower_ci' : conf_int[:,0],
    'upper_ci' : conf_int[:,1]
}, index =test.dropna().index)

fig, ax = plt.subplots(figsize=(14,5))

ax.plot(train['inflation'], color = 'steelblue', label='Train Set', marker='o')
ax.plot(test['inflation'], color = 'tomato', label='Test Set',marker='o')
ax.plot(forecast_df['forecast'], color='mediumseagreen', label='Forecast')
ax.fill_between(forecast_df.index,
                forecast_df['lower_ci'],
                forecast_df['upper_ci'],
                color='tomato',
                alpha=0.3)
ax.axvline(train.index.max(), color='grey', linestyle='--')
ax.set_title('Baseline Model Forecast')

plt.legend()
plt.tight_layout()
plt.savefig('preliminary_plot.png', dpi = 150, bbox_inches = 'tight')
plt.show()

```

SARIMAX Results

```

=====
=====
Dep. Variable:                                y    No. Observations:

```

48

Model: SARIMAX(0, 1, 3)x(1, 0, [], 12) Log Likelihood
-29.406

Date: Mon, 08 Jun 2026 AIC
70.812

Time: 18:14:16 BIC
81.913

Sample: 06-01-2021 HQIC
74.989

- 05-01-2025

Covariance Type: opg

	coef	std err	z	P> z	[0.025	0.975]
intercept	-0.0896	0.143	-0.625	0.532	-0.371	0.191
ma.L1	0.4690	0.167	2.813	0.005	0.142	0.796
ma.L2	-0.1335	0.177	-0.754	0.451	-0.481	0.214
ma.L3	0.2833	0.225	1.259	0.208	-0.158	0.724
ar.S.L12	-0.5904	0.138	-4.280	0.000	-0.861	-0.320
sigma2	0.1765	0.042	4.188	0.000	0.094	0.259

===

Ljung-Box (L1) (Q): 0.04 Jarque-Bera (JB):

0.14

Prob(Q): 0.85 Prob(JB):

0.93

Heteroskedasticity (H): 1.11 Skew:

0.13

Prob(H) (two-sided): 0.84 Kurtosis:

2.98

===

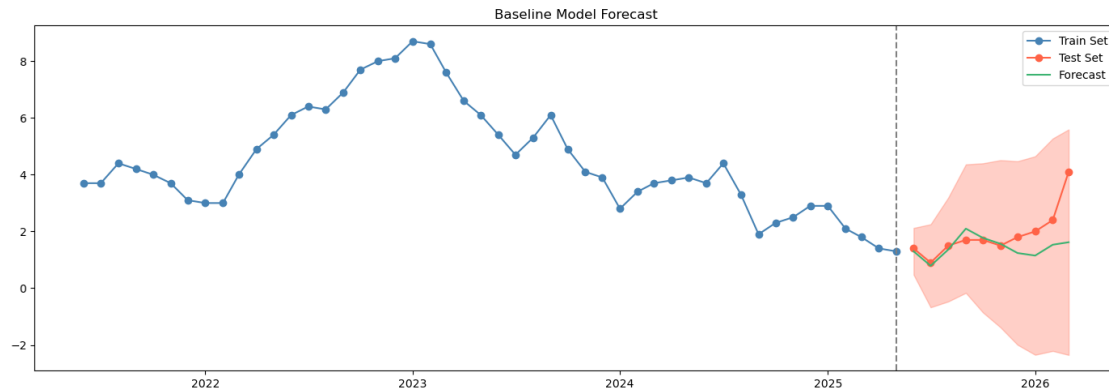
Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

MSE: 0.8155289828913558

RMSE: 0.903066433265768

MAE: 0.5647337379062976



```
[13]: len(test['inflation'].dropna())
```

```
[13]: 10
```

4.3 Feature Engineering for Correction Model

```
[14]: residuals = arima_model.resid()
resid_df = pd.DataFrame(residuals, columns = ['target_residual'])

X_train = train[['sentiment', 'oil_brent']]
y_train = residuals
X_train = X_train.iloc[1:]
y_train = y_train.iloc[1:]

X_test = test[test['inflation'].notna()][['sentiment', 'oil_brent']]
X_future = test[test['inflation'].isna()][['sentiment', 'oil_brent']]
```

4.4 Building Correction Model

```
[15]: rf_model = RandomForestRegressor(
    n_estimators=100,
    max_depth=3,
    min_samples_leaf=5,
    random_state=42
)

rf_model.fit(X_train, y_train)
```

```
[15]: RandomForestRegressor(max_depth=3, min_samples_leaf=5, random_state=42)
```

```
[16]: # RF-Residual Prediction and forecast correction

rf_residual_pred = rf_model.predict(X_test)
sarima_rf_correction = forecast.values+rf_residual_pred
```

```
[17]: xgb_model = XGBRegressor(
        n_estimators = 500,
        max_depth = 3,
        learning_rate = 0.05,
        random_state = 42
    )

xgb_model.fit(X_train,y_train)
```

```
[17]: XGBRegressor(base_score=None, booster=None, callbacks=None,
        colsample_bylevel=None, colsample_bynode=None,
        colsample_bytree=None, device=None, early_stopping_rounds=None,
        enable_categorical=False, eval_metric=None, feature_types=None,
        feature_weights=None, gamma=None, grow_policy=None,
        importance_type=None, interaction_constraints=None,
        learning_rate=0.05, max_bin=None, max_cat_threshold=None,
        max_cat_to_onehot=None, max_delta_step=None, max_depth=3,
        max_leaves=None, min_child_weight=None, missing=nan,
        monotone_constraints=None, multi_strategy=None, n_estimators=500,
        n_jobs=None, num_parallel_tree=None, ...)
```

```
[18]: # XGB Prediction and Forecast Correction
xgb_forecast = xgb_model.predict(X_test)
corrected_xgb_forecast = forecast.values+xgb_forecast
```

```
[19]: # Ensemble Building and Forecast Correction

ensemble_forecast = (xgb_forecast+rf_residual_pred)/2
ensemble_sarima = forecast.values+ensemble_forecast
```

```
[31]: fig, (ax1,ax2,ax3) = plt.subplots(3, 1, figsize=(14,8))

ax1.plot(train['inflation'], color = 'steelblue', label='Train Set', marker='o')
ax1.plot(test['inflation'], color = 'tomato', label='Test Set',marker='o')
ax1.plot(forecast_df['forecast'], color='mediumseagreen', label='Forecast')
ax1.plot(test.dropna().index, sarima_rf_correction, color = 'magenta',
        label='SARIMA-RF Correction')
ax1.fill_between(forecast_df.index,
                forecast_df['lower_ci'],
                forecast_df['upper_ci'],
                color='tomato',
                alpha=0.3)
ax1.axvline(train.index.max(), color='grey', linestyle='--')
ax1.legend()
ax1.set_title('SARIMA vs SARIMA-RF')

ax2.plot(train['inflation'], color = 'steelblue', label='Train Set', marker='o')
```

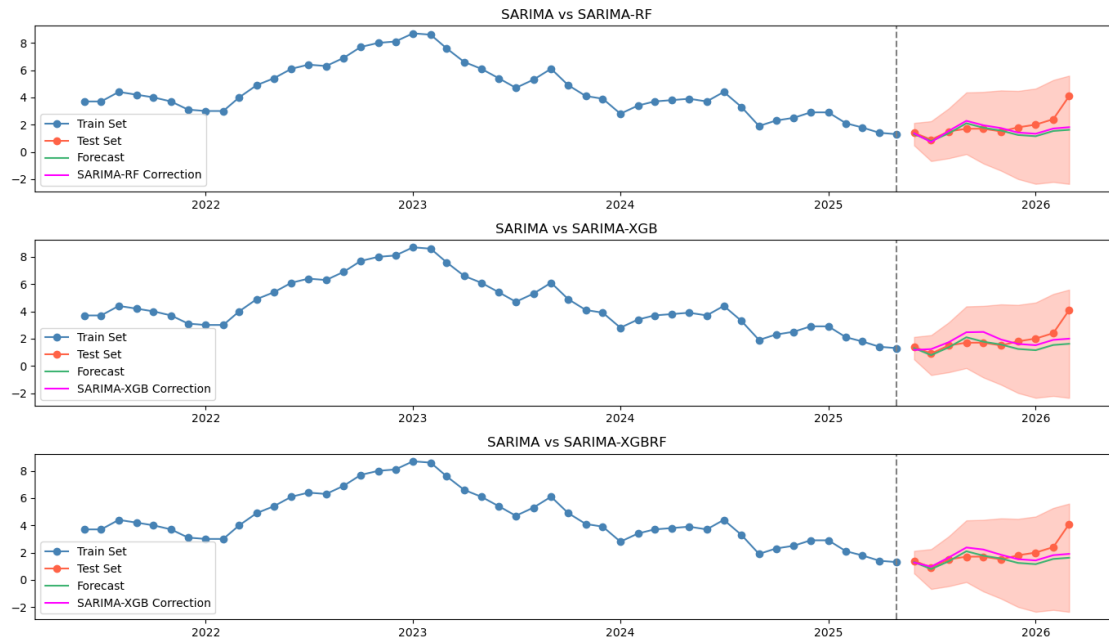
```

ax2.plot(test['inflation'], color = 'tomato', label='Test Set',marker='o')
ax2.plot(forecast_df['forecast'], color='mediumseagreen', label='Forecast')
ax2.plot(test.dropna().index, corrected_xgb_forecast, color = 'magenta',□
    ↪label='SARIMA-XGB Correction')
ax2.fill_between(forecast_df.index,
                 forecast_df['lower_ci'],
                 forecast_df['upper_ci'],
                 color='tomato',
                 alpha=0.3)
ax2.axvline(train.index.max(), color='grey', linestyle='--')
ax2.legend()
ax2.set_title('SARIMA vs SARIMA-XGB')

ax3.plot(train['inflation'], color = 'steelblue', label='Train Set', marker='o')
ax3.plot(test['inflation'], color = 'tomato', label='Test Set',marker='o')
ax3.plot(forecast_df['forecast'], color='mediumseagreen', label='Forecast')
ax3.plot(test.dropna().index, ensemble_sarima, color = 'magenta',□
    ↪label='SARIMA-XGB Correction')
ax3.fill_between(forecast_df.index,
                 forecast_df['lower_ci'],
                 forecast_df['upper_ci'],
                 color='tomato',
                 alpha=0.3)
ax3.axvline(train.index.max(), color='grey', linestyle='--')
ax3.legend()
ax3.set_title('SARIMA vs SARIMA-XGBRF')

plt.tight_layout()
plt.savefig('preliminary_plot1.png', dpi = 150, bbox_inches = 'tight')
plt.show()

```



```
[21]: print(f'SARIMA Baseline Benchmark:')
err_metrics(test['inflation'].dropna(), forecast.values)
print('')
print(f'SARIMA-RF Benchmark:')
err_metrics(test['inflation'].dropna(), sarima_rf_correction)
print('')
print(f'SARIMA-XGB Benchmark:')
err_metrics(test['inflation'].dropna(), corrected_xgb_forecast)
print('')
print(f'SARIMA-XGBRF Benchmark:')
err_metrics(test['inflation'].dropna(), ensemble_sarima)
```

SARIMA Baseline Benchmark:

MSE: 0.8155289828913558

RMSE: 0.903066433265768

MAE: 0.5647337379062976

SARIMA-RF Benchmark:

MSE: 0.6796783017924615

RMSE: 0.8244260438586724

MAE: 0.5367959366646491

SARIMA-XGB Benchmark:

MSE: 0.6566770903987337

RMSE: 0.8103561503430042

MAE: 0.606305249890314

SARIMA-XGBRF Benchmark:
MSE: 0.6471385412718972
RMSE: 0.804449216092537
MAE: 0.5527403376205546

[21]: {None}

1.4.1 5. Final Modeling and Conclusion

5.1 Full Dataset Modeling

```
[22]: # Refitting the Fulldataset to SARIMA with the same orders as before
inflation = df['inflation'].dropna()
inflation.index.freq = 'MS'
final_sarima = SARIMAX(
    inflation,
    order = arima_model.order,
    seasonal_order=arima_model.seasonal_order)

final_sarima_fit = final_sarima.fit(dispatch=False)
```

```
[23]: # Forecasting two observations ahead
final_forecast_obj = final_sarima_fit.get_forecast(steps=2)
final_forecast = final_forecast_obj.predicted_mean
final_conf_int = final_forecast_obj.conf_int()
```

```
[24]: # Getting the residual of final SARIMA model and setting variables for
      ↪ exogenous features
final_resid = final_sarima_fit.resid

final_X_train = df[df['inflation'].notna()][['sentiment', 'oil_brent']]
final_y_train = final_resid
final_X_train = final_X_train.iloc[1:]
final_y_train = final_y_train.iloc[1:]
```

```
[25]: # Refitting to RF model
final_rf = rf_model.fit(final_X_train, final_y_train)
# RF forecast
final_rf_forecast = final_rf.predict(X_future)
```

```
[26]: final_xgb = xgb_model.fit(final_X_train, final_y_train)
final_xgb_forecast = final_xgb.predict(X_future)
```

```
[27]: final_ensemble_correction = final_forecast +
      ↪ (final_rf_forecast + final_xgb_forecast) / 2
```

```
[33]: last_date = df['inflation'].dropna().index[-1]
final_forecast_index = pd.date_range(start=last_date, periods=3, freq='MS')[1:]
```

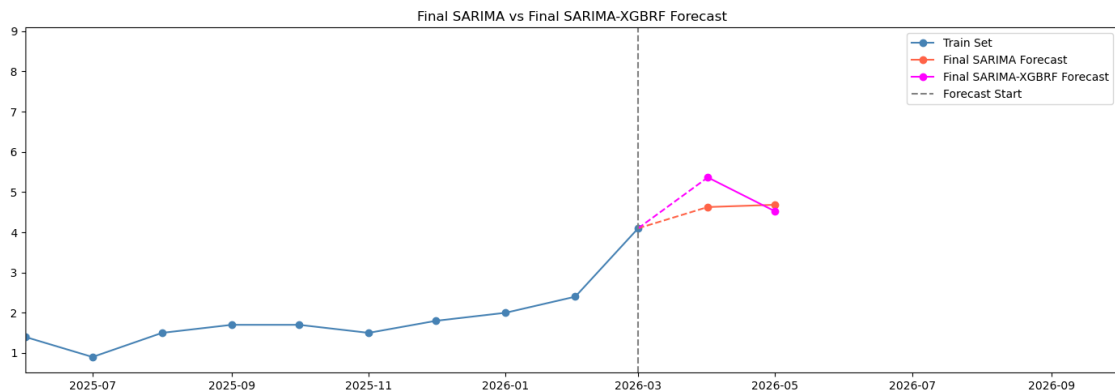
```

fig, ax = plt.subplots(figsize=(14,5))
ax.set_xlim(pd.Timestamp('2025-06-01'), pd.Timestamp('2026-10-01'))
ax.plot(df['inflation'].dropna(), color = 'steelblue', label='Train Set',
        marker='o')
ax.plot(final_forecast_index,final_forecast.values, color='tomato',
        label='Final SARIMA Forecast', marker='o')
ax.plot(final_forecast_index,final_ensemble_correction, color='magenta',
        label='Final SARIMA-XGBRF Forecast', marker='o')
# Get last observed point
last_obs_date = inflation.index[-1]
last_obs_value = inflation.iloc[-1]

# Connect observation to forecast
ax.plot([last_obs_date, final_forecast_index[0]],
        [last_obs_value, final_forecast.values[0]],
        color='tomato', linestyle='--')

ax.plot([last_obs_date, final_forecast_index[0]],
        [last_obs_value, final_ensemble_correction.values[0]],
        color='magenta', linestyle='--')
ax.axvline(last_date, color='grey', linestyle='--', label='Forecast Start')
ax.set_title('Final SARIMA vs Final SARIMA-XGBRF Forecast')
plt.legend()
plt.savefig('Final Forecast_plot.png', dpi = 150, bbox_inches = 'tight')
plt.tight_layout()
plt.show()

```



[29]: # Forecast Values

```
print("Final Forecast:")
```

```
print(f"April 2026 - SARIMA: {final_forecast.values[0]:.2f}, Ensemble:␣  
↪{final_ensemble_correction.values[0]:.2f}")  
print(f"May 2026 - SARIMA: {final_forecast.values[1]:.2f}, Ensemble:␣  
↪{final_ensemble_correction.values[1]:.2f}")
```

Final Forecast:

April 2026 - SARIMA: 4.63, Ensemble: 5.37

May 2026 - SARIMA: 4.68, Ensemble: 4.52

1.4.2 Conclusion and Findings

- This project investigated whether macroeconomic indicators — specifically Brent crude oil prices and financial news sentiment — could meaningfully improve short-term Philippine inflation forecasting beyond a univariate SARIMA baseline.
- Correlation analysis confirmed that both features carry a positive leading relationship with inflation: oil prices at lag 2 ($r = 0.67$) and finBERT sentiment scores at lag 2 ($r = 0.44$). USD-PHP exchange rate was initially hypothesized as a third feature, but correlation analysis using verifiable BSP and yfinance data produced weak results ($r < 0.25$ across all lags), suggesting that BSP monetary intervention may be suppressing the direct pass-through effect between exchange rate movements and inflation. The feature was dropped to maintain data integrity.
- The hybrid SARIMA-XGBRF ensemble model outperformed the standalone SARIMA baseline with an RMSE of 0.804 compared to 0.903, representing a 20.73% improvement in forecasting accuracy. All three correction models — Random Forest, XGBoost, and their ensemble — consistently beat the baseline, confirming that exogenous macroeconomic features add meaningful predictive signal beyond what ARIMA can capture alone.
- The final 2-month ahead forecast using real observed exogenous values projects April 2026 inflation at 5.37% and May 2026 at 4.52%, suggesting a short-term spike followed by partial moderation. Both figures remain elevated relative to the low inflation environment observed in mid-2025.
- Limitations: The model is constrained by monthly data aggregation — oil prices and sentiment exhibit high daily volatility that is lost when averaged to monthly frequency. The correction layer is further limited by the near-white-noise behavior of SARIMA residuals, leaving minimal learnable signal for the ensemble models.
- Finally, the 2-month forecast horizon is bounded by the availability of leading exogenous data; forecasting beyond this window would require separate models for the exogenous variables themselves.